

RETAIL CUSTOMER 360 DATA ENGINEERING
PROJECT
WITH ADF AND DATABRICKS

BY :

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OBJECTIVE

The objective of this Retail Customer 360 Data Engineering Pipeline is to process retail data from multiple source systems in a reliable, scalable, and controlled manner while ensuring data accuracy, consistency, and availability for analytics and reporting. The pipeline is designed to process only new or changed data using incremental loading techniques, preventing duplicate processing and improving overall efficiency.

The project ingests retail datasets such as customers, products, stores, inventory, and sales into Azure Data Lake Storage Gen2 and processes them through Raw, Bronze, Silver, and Gold layers using Databricks and Delta Lake. Data is validated, cleaned, transformed, and structured into analytics-ready datasets for downstream business intelligence and reporting in Power BI.

The pipeline is built using production-oriented practices such as idempotent execution, Delta Lake merge operations, rejected record handling, watermark-based incremental processing, and audit-friendly data organization to ensure reliability and maintainability.

KEY CAPABILITIES

- Process only new or unprocessed retail data using incremental loading
- Handle duplicate and invalid records through validation and rejected-data handling
- Maintain data consistency using Delta Lake merge operations
- Implement Medallion Architecture (Raw-Bronze -Silver-Gold)
- Support rerunnable and idempotent pipeline execution
- Perform data cleaning, deduplication, and transformation using PySpark
- Enable scalable storage and processing using ADLS Gen2 and Databricks
- Create analytics-ready reporting tables for Power BI dashboards
- Support auditability and operational monitoring through structured data processing
- Ensure scalability for growing retail transaction and customer datasets

SCOPE

This pipeline covers ingestion and processing of multiple retail datasets from source systems into the cloud data platform. It includes movement of data across Raw, Bronze, Silver, and Gold layers along with validation, transformation, cleansing, deduplication, and rejected-record handling.

The pipeline processes structured retail datasets and ensures only new or unprocessed data is moved forward using incremental loading techniques. It supports end-to-end processing from ingestion to analytics-ready datasets for reporting and business intelligence.

Key Scope Points

- Covers multiple retail datasets such as customers, products, stores, inventory, and sales
- Includes data ingestion from source systems into Azure Data Lake Storage Gen2
- Processes data through Raw, Bronze, Silver, and Gold layers
- Applies data validation, cleansing, standardization, and transformation rules
- Handles duplicate and invalid records through rejected-data processing
- Maintains reliable processing using Delta Lake merge operations
- Supports incremental data processing using watermark logic
- Supports rerunnable and idempotent execution without duplication
- Creates analytics-ready reporting datasets for Power BI dashboards
- Enables scalable cloud-based processing using Azure Databricks and Delta Lake

DATA FLOW

The pipeline ingests retail datasets from on-premises SQL systems and source batch files into the Raw layer in Azure Data Lake Storage Gen2 using batch-based ingestion. Source files are stored in their original format without modification to preserve raw data integrity and support auditability.

Data is then moved into the Bronze layer, where it is converted into a standardized parquet/delta format and organized for scalable downstream processing. Incremental processing logic is applied to ensure that only new or unprocessed data is handled during each pipeline execution.

From the Bronze layer, data is validated, cleaned, standardized, and transformed using PySpark within Azure Databricks before being loaded into the Silver layer. In this stage, invalid and malformed records are separated into rejected datasets while valid records are deduplicated and refined for business use.

The refined Silver-layer datasets are further transformed into Gold-layer dimensional and fact tables optimized for analytics and reporting. These reporting tables are consumed by Power BI dashboards for business intelligence and visualization.

The pipeline maintains audit-friendly processing across all stages and is designed using rerunnable and idempotent execution logic. It also handles no-data scenarios gracefully without causing pipeline failures, ensuring reliable production-oriented execution.

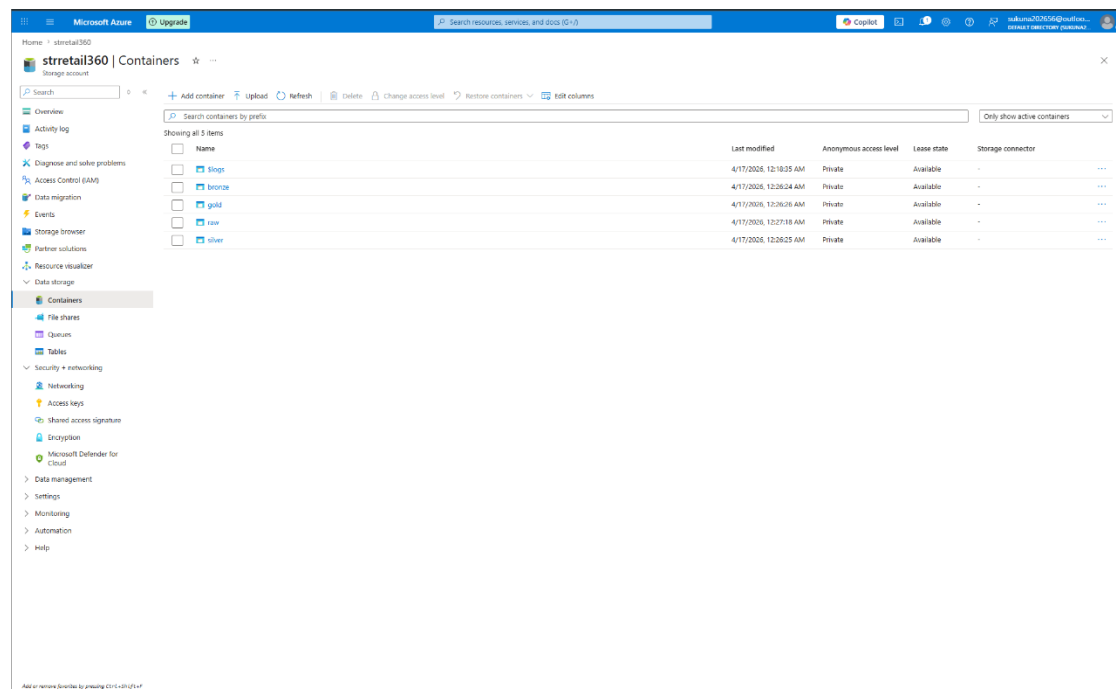
ARCHITECTURE

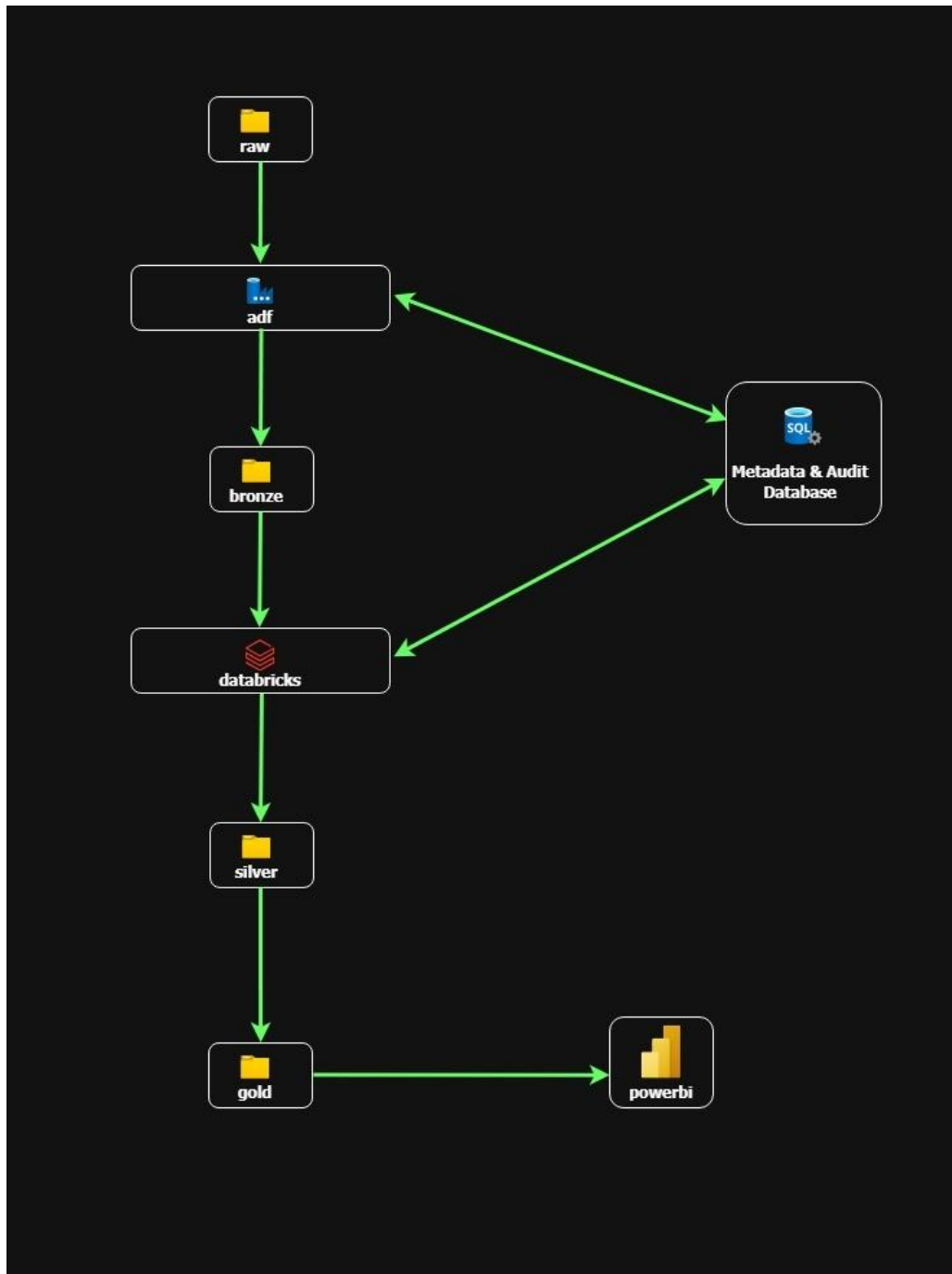
The pipeline is built using Azure Data Factory for orchestration, Azure Data Lake Storage Gen2 for scalable cloud storage, Azure Databricks for distributed data processing, and Delta Lake for reliable incremental data management. Data is ingested from on-premise SQL Server systems and source batch files into ADLS Gen2 and processed through Raw, Bronze, Silver, and Gold layers following the Medallion Architecture approach.

Azure Data Factory pipelines manage end-to-end orchestration, including ingestion, incremental processing, notebook execution, and pipeline control flow. Azure Databricks notebooks perform data validation, cleansing, standardization, deduplication, rejected-record handling, and business transformations using PySpark.

The Bronze layer stores standardized source data, the Silver layer contains cleansed and validated business-ready datasets, and the Gold layer contains dimensional and fact tables optimized for analytics and reporting. Delta Lake merge operations are implemented to support rerunnable and idempotent processing while maintaining data consistency across all layers.

The architecture follows a modular and scalable design where each dataset is processed independently through reusable logic controlled by orchestration pipelines, improving maintainability, operational monitoring, and troubleshooting efficiency across the platform.





PIPELINE DESIGN & CONTROL TABLE

The pipeline is designed using a metadata-driven and modular architecture to support scalable and maintainable processing of multiple retail datasets. Azure Data Factory orchestrates ingestion and notebook execution while Azure Databricks performs data transformation, validation, cleansing, deduplication, and Delta Lake merge processing.

A centralized control table is used to manage dataset-level processing information such as source paths, watermark columns, load status, and last processed timestamps. This enables incremental processing, rerunnable execution, and controlled orchestration across all datasets including customers, products, stores, inventory, and sales.

The pipeline processes data through Raw, Bronze, Silver, and Gold layers following the Medallion Architecture approach. Incremental loading logic ensures that only new or unprocessed data is moved through the pipeline during each execution cycle, improving efficiency and preventing duplicate processing.

The design supports modular execution, audit-friendly processing, structured data organization, rejected-record handling, and scalable cloud-based analytics integration with Power BI.

Factory Resources	
Filter resources by name	
Pipelines	16
clean_tables	2
master_pipeline	3
master_trigger	1
sub_bronze_to_silver	1
sub_raw_to_bronze	1
pipelines_architecture	11
bronze_to_silver_transform	5
pl_customers_bronze_to_silver	
pl_inventory_bronze_to_silver	
pl_products_bronze_to_silver	
pl_sales_bronze_to_silver	
pl_stores_bronze_to_silver	
raw_to_bronze_pipelines	5
pl_raw_to_bronze_customer	
pl_raw_to_bronze_inventory	
pl_raw_to_bronze_product	
pl_raw_to_bronze_sales	
pl_raw_to_bronze_store	
silver_to_gold_aggregate	1
pl_gold_aggregate	

control_table								
Data preview - control_table								
Showing 1000 rows								
	ABC pipeline_name	ABC source_name	last_watermark_value	current_watermark_value	ABC load_type	ABC status	last_run_time	ABC run_id
1	pl_bronze_to_silver_customer	customers	2026-05-06T21:13:01.7330000	2026-05-07T02:15:38.4100000	incremental	INIT	2026-05-07T02:15:38.4100000	57c172c4-aa1a-4d1f-a43f-497adc7...
2	pl_bronze_to_silver_product	products	2026-05-06T21:15:51.8630000	2026-05-07T02:19:56.7770000	incremental	INIT	2026-05-07T02:19:56.7770000	bed29c8f-dc5b-4c86-b3d5-3141c4...
3	pl_bronze_to_silver_store	stores	2026-05-06T21:19:18.8930000	2026-05-07T02:25:29.2330000	incremental	INIT	2026-05-07T02:25:29.2330000	a9dec2e8-be38-45cb-be35-9129ec...
4	pl_bronze_to_silver_inventory	inventory	2026-05-06T21:23:29.3970000	2026-05-07T02:32:28.0330000	incremental	INIT	2026-05-07T02:32:28.0330000	6f3f7ffa-e7a4-4ddc-9db7-49da872...
5	pl_customers_bronze_to_silver	customers	2000-01-01T00:00:00.0000000	2026-05-06T21:11:58.0000000	incremental	Success	2026-05-06T21:37:09.8070000	5a1075cd-fb1b-49fe-8bfd-6ffefa14...
6	pl_bronze_to_silver_sales	sales	2026-05-06T21:26:29.8500000	2026-05-07T02:37:03.2370000	incremental	INIT	2026-05-07T02:37:03.2370000	1a78508e-6621-44a1-89db-73bcc2...
7	pl_inventory_bronze_to_silver	inventory	2000-01-01T00:00:00.0000000	2026-05-06T21:20:58.0000000	incremental	Success	2026-05-06T21:38:37.9670000	2103be89-46ba-4511-be0f-739e87...
8	pl_products_bronze_to_silver	products	2000-01-01T00:00:00.0000000	2026-05-06T21:14:29.0000000	incremental	Success	2026-05-06T21:40:07.1500000	38d7394d-1454-4aa6-a32e-33a432...
9	pl_sales_bronze_to_silver	sales	2000-01-01T00:00:00.0000000	2026-05-06T21:24:53.0000000	incremental	Success	2026-05-06T21:41:32.1570000	fe2888d5-96cb-405c-aa14-88fcd29...
10	pl_stores_bronze_to_silver	stores	2000-01-01T00:00:00.0000000	2026-05-06T21:17:52.0000000	incremental	Success	2026-05-06T21:42:50.1000000	d02973e6-4f2a-424a-86c3-3cd3c1c...

ControlTable: Tracks last processed timestamp for each dataset to enable incremental loading

audit_table												
Data preview - audit_table												
Showing 1000 rows												
	123 audit_id	ABC run_id	ABC pipeline_name	ABC source_name	ABC activity_name	start_time	end_time	ABC status	ABC error_message	123 rows_read	123 rows_written	
1	1	fe1bb918-baea-4227-9c...	pl_raw_to_bronze_custo...	customers	cp_raw_to_bronze	2026-05-06T21:12:43.76...	2026-05-06T21:12:43.76...	SUCCESS	NULL	0	0	
2	2	4a5f2a2c-c8e9-40a1-b1...	pl_raw_to_bronze_product	products	cp_raw_to_bronze	2026-05-06T21:15:22.35...	2026-05-06T21:15:22.35...	SUCCESS	NULL	0	0	
3	3	a963ae63-5edb-4e2a-91...	pl_raw_to_bronze_store	stores	cp_raw_to_bronze	2026-05-06T21:18:51.36...	2026-05-06T21:18:51.36...	SUCCESS	NULL	0	0	
4	4	15d366e8-953b-457a-9c...	pl_raw_to_bronze_invent...	inventory	cp_raw_to_bronze	2026-05-06T21:23:05.77...	2026-05-06T21:23:05.77...	SUCCESS	NULL	0	0	
5	5	ec5c8760-9966-42d2-9f...	pl_raw_to_bronze_sales	sales	cp_raw_to_bronze	2026-05-06T21:25:53.25...	2026-05-06T21:25:53.25...	SUCCESS	NULL	0	0	
6	6	5a1075cd-fb1b-49fe-8bf...	pl_customers_bronze_to...	customers	sp_log_pipeline_start	2026-05-06T21:27:08.00...	2026-05-06T21:37:18.28...	Success	NULL	NULL	NULL	
7	7	2103be89-46ba-4511-b...	pl_inventory_bronze_to...	inventory	sp_log_pipeline_start	2026-05-06T21:37:36.95...	2026-05-06T21:38:47.44...	Success	NULL	NULL	NULL	
8	8	38d7394d-1454-4aa6-a3...	pl_products_bronze_to_s...	products	sp_log_pipeline_start	2026-05-06T21:39:06.13...	2026-05-06T21:40:14.48...	Success	NULL	NULL	NULL	
9	9	fe2888d5-96cb-405c-aa...	pl_sales_bronze_to_silv...	sales	sp_log_pipeline_start	2026-05-06T21:40:30.16...	2026-05-06T21:41:39.52...	Success	NULL	NULL	NULL	
10	10	d02973e6-4f2a-424a-86...	pl_stores_bronze_to_silv...	stores	sp_log_pipeline_start	2026-05-06T21:41:56.08...	2026-05-06T21:42:57.26...	Success	NULL	NULL	NULL	
11	11	57c172c4-aa1a-4d1f-a4...	pl_raw_to_bronze_custo...	customers	cp_raw_to_bronze	2026-05-07T02:14:06.11...	2026-05-07T02:14:06.11...	SUCCESS	NULL	0	0	
12	12	57c172c4-aa1a-4d1f-a4...	pl_raw_to_bronze_custo...	customers	cp_raw_to_bronze	2026-05-07T02:15:13.44...	2026-05-07T02:15:13.44...	SUCCESS	NULL	0	0	
13	13	bed29c8f-dc5b-4c86-b3...	pl_raw_to_bronze_product	products	cp_raw_to_bronze	2026-05-07T02:18:24.83...	2026-05-07T02:18:24.83...	SUCCESS	NULL	0	0	
14	14	bed29c8f-dc5b-4c86-b3...	pl_raw_to_bronze_product	products	cp_raw_to_bronze	2026-05-07T02:19:37.11...	2026-05-07T02:19:37.11...	SUCCESS	NULL	0	0	
15	15	a9dec2e8-be38-45cb-be...	pl_raw_to_bronze_store	stores	cp_raw_to_bronze	2026-05-07T02:23:52.03...	2026-05-07T02:23:52.03...	SUCCESS	NULL	0	0	
16	16	a9dec2e8-be38-45cb-be...	pl_raw_to_bronze_store	stores	cp_raw_to_bronze	2026-05-07T02:25:09.86...	2026-05-07T02:25:09.86...	SUCCESS	NULL	0	0	
17	17	6f3f7ffa-e7a4-4ddc-9db...	pl_raw_to_bronze_invent...	inventory	cp_raw_to_bronze	2026-05-07T02:27:54.50...	2026-05-07T02:27:54.50...	SUCCESS	NULL	0	0	
18	18	6f3f7ffa-e7a4-4ddc-9db...	pl_raw_to_bronze_invent...	inventory	cp_raw_to_bronze	2026-05-07T02:32:01.67...	2026-05-07T02:32:01.67...	SUCCESS	NULL	0	0	
19	19	1a78508e-6621-44a1-89...	pl_raw_to_bronze_sales	sales	cp_raw_to_bronze	2026-05-07T02:34:44.00...	2026-05-07T02:34:44.00...	SUCCESS	NULL	0	0	
20	20	1a78508e-6621-44a1-89...	pl_raw_to_bronze_sales	sales	cp_raw_to_bronze	2026-05-07T02:36:16.35...	2026-05-07T02:36:16.35...	SUCCESS	NULL	0	0	
21	21	NULL	pl_raw_to_bronze_custo...	customers	fe_loop_files	2026-05-07T03:20:31.78...	2026-05-07T03:20:31.78...	NO_NEW_FILES	NULL	NULL	NULL	
22	22	NULL	pl_raw_to_bronze_product	products	fe_loop_files	2026-05-07T03:23:58.17...	2026-05-07T03:23:58.17...	NO_NEW_FILES	NULL	NULL	NULL	
23	23	NULL	pl_raw_to_bronze_store	stores	fe_loop_files	2026-05-07T03:28:51.81...	2026-05-07T03:28:51.81...	NO_NEW_FILES	NULL	NULL	NULL	
Succeeded (0 sec 229 ms)												
											Columns: 11 Rows: 25	

AuditLog: Captures pipeline execution details including status, layer, and timing for monitoring and debugging

file_tracking_table								
Data preview - file_tracking_table								
Showing 1000 rows								
	123 file_id	ABC file_name	ABC file_path	file_date	o/h processed_flag	processed_time	ABC run_id	ABC layer
1	1	customer_raw_batch1.csv	customers/customer_raw_batch1.csv	2026-05-06	True	2026-05-06T21:12:27.6870000	fe1bb918-baea-4227-9cb7-0d9269...	bronze
2	2	product_raw_batch1.csv	products/product_raw_batch1.csv	2026-05-06	True	2026-05-06T21:15:08.2230000	4a5f2a2c-c8e9-40a1-b152-6d0ae6...	bronze
3	3	store_raw_batch1.csv	stores/store_raw_batch1.csv	2026-05-06	True	2026-05-06T21:18:36.2830000	a963ae63-5edb-4e2a-9123-30f758...	bronze
4	4	inventory_raw_batch1.json	inventory/inventory_raw_batch1.json	2026-05-06	True	2026-05-06T21:22:22.4400000	15d366e8-953b-457a-9cd9-c6e23f...	bronze
5	5	sales_raw_batch1.json	sales/sales_raw_batch1.json	2026-05-06	True	2026-05-06T21:25:23.8200000	ec5c8760-9966-42d2-9f07-3323c28...	bronze
6	6	customer_raw_batch2.csv	customers/customer_raw_batch2.csv	2026-05-07	True	2026-05-07T02:13:47.2830000	57c172c4-aa1a-4d1f-a43f-497adc7...	bronze
7	7	customer_raw_batch3.csv	customers/customer_raw_batch3.csv	2026-05-07	True	2026-05-07T02:14:57.4870000	57c172c4-aa1a-4d1f-a43f-497adc7...	bronze
8	8	product_raw_batch2.csv	products/product_raw_batch2.csv	2026-05-07	True	2026-05-07T02:18:01.3470000	bed29c8f-dc5b-4c86-b3d5-3141c4...	bronze
9	9	product_raw_batch3.csv	products/product_raw_batch3.csv	2026-05-07	True	2026-05-07T02:19:21.6900000	bed29c8f-dc5b-4c86-b3d5-3141c4...	bronze
10	10	store_raw_batch2.csv	stores/store_raw_batch2.csv	2026-05-07	True	2026-05-07T02:23:30.3600000	a9dec2e8-be38-45cb-be35-9129ec...	bronze
11	11	store_raw_batch3.csv	stores/store_raw_batch3.csv	2026-05-07	True	2026-05-07T02:24:57.1270000	a9dec2e8-be38-45cb-be35-9129ec...	bronze
12	12	inventory_raw_batch2.json	inventory/inventory_raw_batch2.json	2026-05-07	True	2026-05-07T02:27:42.1000000	6f3f7ffa-e7a4-4ddc-9db7-49da872...	bronze
13	13	inventory_raw_batch3.json	inventory/inventory_raw_batch3.json	2026-05-07	True	2026-05-07T02:31:42.4600000	6f3f7ffa-e7a4-4ddc-9db7-49da872...	bronze
14	14	sales_raw_batch2.json	sales/sales_raw_batch2.json	2026-05-07	True	2026-05-07T02:34:31.2570000	1a78508e-6621-44a1-89db-73bcc2...	bronze
15	15	sales_raw_batch3.json	sales/sales_raw_batch3.json	2026-05-07	True	2026-05-07T02:35:52.1500000	1a78508e-6621-44a1-89db-73bcc2...	bronze

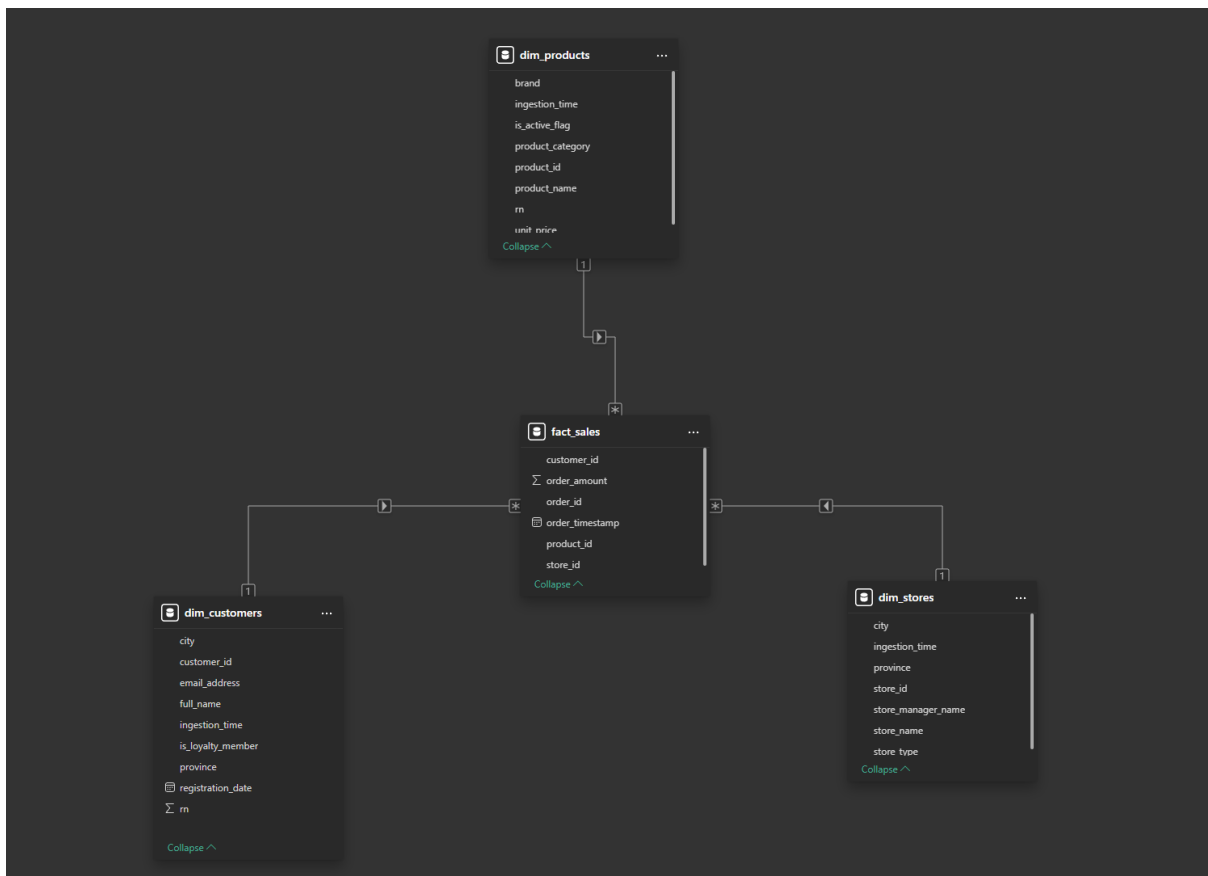
FilesTable : manages file-level tracking and status to support dynamic, idempotent pipeline execution.

DATASET

The project uses synthetic retail datasets generated to simulate a realistic retail business environment without exposing sensitive business information. The datasets include Customers, Products, Stores, Inventory, and Sales, each representing core retail business domains. These datasets are interconnected using primary and foreign keys such as `customer_id`, `product_id`, and `store_id` to model real-world retail relationships and transactions.

The datasets contain intentionally introduced inconsistencies such as duplicate records, null values, malformed dates, invalid email formats, and incorrect categorical values to simulate real-world data quality challenges. This enabled implementation of validation, cleansing, deduplication, rejected-record handling, and transformation logic within the pipeline.

All datasets are ingested into the Raw layer, standardized within the Bronze layer, validated and cleansed in the Silver layer, and transformed into analytics-ready dimensional and fact tables in the Gold layer to support Power BI dashboards and business reporting.



Retail Customer 360 – Star Schema Data Model in Power BI

TECHNOLOGIES USED

COMPONENT	TECHNOLOGY
Cloud Platform	Microsoft Azure
Data Orchestration	Azure Data Factory (ADF)
Data Storage	Azure Data Lake Storage Gen2 (ADLS Gen2)
Data Processing	Azure Databricks
Programming Language	Python / PySpark
Storage Format	Parquet / Delta Lake
Data Transformation	PySpark DataFrames
Incremental Processing	Watermark Logic
Data Modeling	Star Schema
Reporting & Visualization	Power BI
Version Control	Git / GitHub
Architecture Pattern	Medallion Architecture
Data Integration	Batch Processing
Data Quality Handling	Validation & Rejected Record Processing

DATA VALIDATION & TRANSFORMATION

The pipeline implements multiple data validation and transformation rules within the Silver layer using PySpark to improve data quality, consistency, and reliability before downstream analytics processing.

Validation logic is applied to identify invalid, malformed, duplicate, and incomplete records. Invalid records are separated into rejected datasets for auditability and controlled processing while valid records are standardized and transformed into business-ready datasets.

Validation & Transformation Activities

- Removal of duplicate records using deduplication logic

- Validation of mandatory fields such as customer_id and product_id
- Standardization of text columns using trimming and case formatting
- Email format validation using regular expressions
- Date parsing and standardization into consistent formats
- Normalization of categorical fields such as province and loyalty indicators
- Handling of null and malformed values
- Separation of invalid records into rejected-data storage
- Delta Lake merge operations for incremental upsert processing
- Creation of analytics-ready dimensional and fact tables for reporting

The transformation layer ensures that only validated and curated data is propagated into downstream Gold-layer reporting datasets and Power BI dashboards.

2 hours ago (3s) 10

```
path = "abfss://silver@strretail1360.dfs.core.windows.net/customers/customers_silver/"
df = spark.read.format("delta").load(path)

df.display()

print("Row Count:", df.count())
```

(4) Spark Jobs

> df: pyspark.sql.connect.dataframe.DataFrame = [customer_id: string, full_name: string ... 7 more fields]

	customer_id	full_name	email_address	city	province	registration_date	is_loyalty_member	ingestion_time	rn
1	CUST-26-000001	Nicole White	brianbell@example.com	Iqaluit	NU	2020-09-21	null	2026-05-06T21:11:58.367844...	1
2	CUST-26-000002	Cristian Jacobs	jeffwallace@example.com	Kingston	ON	2025-11-11	null	2026-05-06T21:11:58.367844...	1
3	CUST-26-000003	Dr. Connie Smith Dds	rossanthony@example.net	Cochrane	XX	2025-12-03	null	2026-05-06T21:11:58.367844...	1
4	CUST-26-000006	Gary White	johnwebb@example.net	Portage La Prairie	ON	2021-05-20	null	2026-05-06T21:11:58.367844...	1
5	CUST-26-000007	James Booker	gcruz@example.net	Grand Falls-windsor	NL	2021-04-26	null	2026-05-06T21:11:58.367844...	1
6	CUST-26-000008	Kathryn Bond	fjones@example.com	Richmond Hill	null	2025-02-10	null	2026-05-06T21:11:58.367844...	1
7	CUST-26-000009	Alexander Bailey	kimfields@example.org	Souris	PE	2023-02-17	false	2026-05-06T21:11:58.367844...	1
8	CUST-26-000012	Jesus Barnes	kscott@example.com	Marystown	NL	2017-02-20	null	2026-05-06T21:11:58.367844...	1
9	CUST-26-000013	Megan Smith	xbolton@example.org	Fort Smith	NT	2015-07-23	null	2026-05-06T21:11:58.367844...	1
10	CUST-26-000014	Veronica Martinez	adamsjohn@example.org	The Pas	MB	2017-09-01	null	2026-05-06T21:11:58.367844...	1
11	CUST-26-000018	Frank Jones	amy58@example.net	Okotoks	AB	2016-05-24	null	2026-05-06T21:11:58.367844...	1
12	CUST-26-000020	Jordan Wang	kyle50@example.org	Igloolik	NU	2023-06-20	null	2026-05-06T21:11:58.367844...	1
13	CUST-26-000021	Norma Moore	emily93@example.org	Morden	MB	2024-04-08	null	2026-05-06T21:11:58.367844...	1
14	CUST-26-000022	Sharon Yu	matthewrobbins@example.org	Souris	PE	2022-09-17	null	2026-05-06T21:11:58.367844...	1
15	CUST-26-000024	David Sandoval	gutierrezsheena@example.com	Rouyn-noranda	XX	2022-09-02	null	2026-05-06T21:11:58.367844...	1

53 rows | 2.65s runtime

Row Count: 53

Silver Layer Inventory Transformation – Databricks Notebook

2 hours ago (2s) 10

```

silver_path = "abfss://silver@strretail360.dfs.core.windows.net/stores/stores_silver"

silver_df = spark.read.format("delta").load(silver_path)

silver_df.display()

```

(2) Spark Jobs

silver_df: pyspark.sql.connect.dataframe.DataFrame = [store_id: string, store_name: string ... 5 more fields]

	store_id	store_name	store_type	city	province	store_manager_name	ingestion_time
1	STR-26-000002	Martin Ltd Mall	mall	null	null	Cathy Roberts	2026-05-06T21:17:52.316171...
2	STR-26-000003	Henderson LLC Outlet	outlet	Stratford	PE	Katherine Martin	2026-05-06T21:17:52.316171...
3	STR-26-000005	Martin and Sons Mall	mall	Calgary	QC	Amber Wilson	2026-05-06T21:17:52.316171...
4	STR-26-000006	Obrien, Harding and Butler Outlet	outlet	null	null	Dr. Denise Jones	2026-05-06T21:17:52.316171...
5	STR-26-000007	Erickson-Ashley Outlet	outlet	Teslin	YT	Antonio Rose	2026-05-06T21:17:52.316171...
6	STR-26-000008	Liu Group Mall	mall	Surrey	BC	Joseph Wise	2026-05-06T21:17:52.316171...
7	STR-26-000011	Holmes, Murphy and Duncan Outlet	outlet	Haines Juncti...	null	James Solis	2026-05-06T21:17:52.316171...
8	STR-26-000012	Logan, Mack and White Outlet	outlet	null	NS	Michael Gardner	2026-05-06T21:17:52.316171...
9	STR-26-000013	Stewart, Hall and Gonzalez Outlet	outlet	Vancouver	BC	Chelsea Tran	2026-05-06T21:17:52.316171...
10	STR-26-000014	James-Lewis Outlet	outlet	null	SK	George Ellis	2026-05-06T21:17:52.316171...
11	STR-26-000015	Williams, Waller and Robertson Flagship	flagship	Rankin Inlet	NU	Sharon Ward	2026-05-06T21:17:52.316171...

11 rows | 2.03s runtime

Silver Layer stores Transformation – Databricks Notebook

2 hours ago (2s) 11

```

silver_path = "abfss://silver@strretail360.dfs.core.windows.net/sales/sales_silver"

silver_df = spark.read.format("delta").load(silver_path)

silver_df.display()

```

(2) Spark Jobs

silver_df: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, customer_id: string ... 6 more fields]

	order_id	customer_id	product_id	store_id	order_timestamp	order_amount	sales_channel	ingestion_time
1	ORD-26-000001	CUST-26-000031	PRD-26-000029	STR-26-000003	2018-02-13T13:07:55.000+00...	242.42	online	2026-05-07T02:35:14.416571...
2	ORD-26-000002	CUST-26-000007	PRD-26-000034	STR-26-000009	2017-06-04T10:28:06.000+00...	241.9	online	2026-05-07T02:35:14.416571...
3	ORD-26-000003	CUST-26-000023	PRD-26-000028	STR-26-000003	2016-08-11T18:31:06.000+00...	1498.59	store	2026-05-07T02:35:14.416571...
4	ORD-26-000004	CUST-26-000045	PRD-26-000016	STR-26-000001	2016-07-27T12:38:52.000+00...	55.29	online	2026-05-07T02:35:14.416571...
5	ORD-26-000005	CUST-26-000045	PRD-26-000020	STR-26-000009	2018-05-16T17:11:23.000+00...	27.97	store	2026-05-07T02:35:14.416571...
6	ORD-26-000006	CUST-26-000035	PRD-26-000045	STR-26-000001	2017-03-17T10:49:01.000+00...	37.36	online	2026-05-07T02:35:14.416571...
7	ORD-26-000007	CUST-26-000033	PRD-26-000039	STR-26-000001	2016-09-21T10:01:12.000+00...	1520.86	online	2026-05-07T02:35:14.416571...
8	ORD-26-000008	CUST-26-000020	PRD-26-000029	STR-26-000001	2016-06-28T18:58:48.000+00...	249.33	store	2026-05-07T02:35:14.416571...
9	ORD-26-000009	CUST-26-000055	PRD-26-000051	STR-26-000011	2016-07-13T19:38:37.000+00...	1335.55	store	2026-05-07T02:35:14.416571...
10	ORD-26-000010	CUST-26-000058	PRD-26-000075	STR-26-000007	2018-01-20T18:07:08.000+00...	8.03	store	2026-05-07T02:35:14.416571...
11	ORD-26-000011	CUST-26-000031	PRD-26-000078	STR-26-000009	2016-07-26T13:25:47.000+00...	34.75	online	2026-05-07T02:35:14.416571...
12	ORD-26-000012	CUST-26-000020	PRD-26-000049	STR-26-000014	2016-09-04T18:41:23.000+00...	241.55	online	2026-05-07T02:35:14.416571...
13	ORD-26-000013	CUST-26-000020	PRD-26-000049	STR-26-000003	2017-06-28T20:41:59.000+00...	242.96	store	2026-05-07T02:35:14.416571...
14	ORD-26-000014	CUST-26-000040	PRD-26-000055	STR-26-000003	2018-07-28T17:43:05.000+00...	245.78	store	2026-05-07T02:35:14.416571...
15	ORD-26-000015	CUST-26-000058	PRD-26-000036	STR-26-000002	2017-10-16T17:54:27.000+00...	31.9	store	2026-05-07T02:35:14.416571...

2,721 rows | 1.80s runtime

Silver Layer sales Transformation – Databricks Notebook

INCREMENTAL LOADING & WATERMARK PROCESSING

The pipeline implements incremental loading techniques to ensure that only new or unprocessed retail data is processed during each execution cycle. This approach improves processing efficiency, reduces unnecessary computation, and prevents duplicate data movement across the platform.

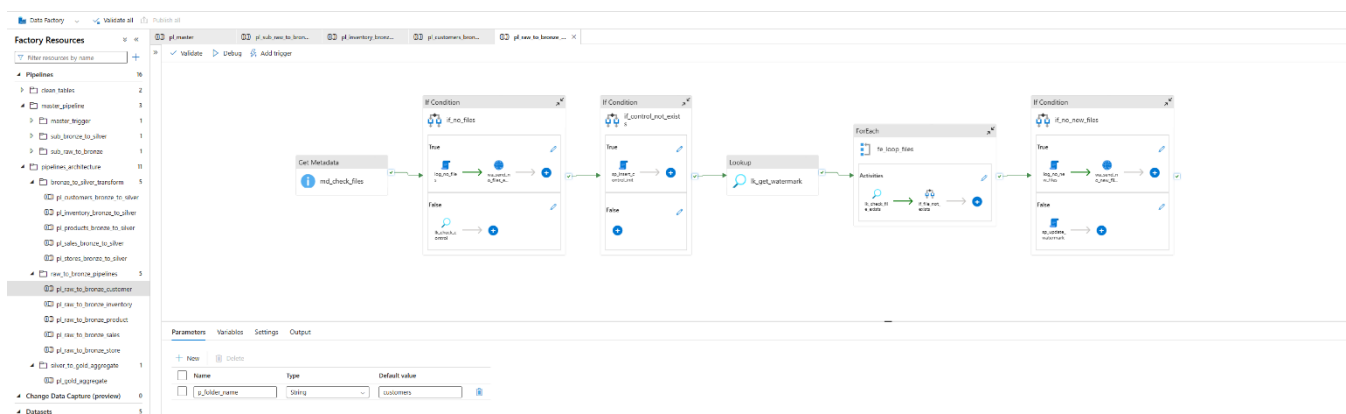
A centralized control table is used to maintain dataset-level watermark information such as last processed timestamps, load status, and execution metadata. During pipeline execution, the system compares incoming data against stored watermark values to identify new records eligible for processing.

Delta Lake merge operations are implemented within the Silver and Gold layers to support controlled upsert functionality. Existing records are updated while new records are inserted without duplicating previously processed data.

The incremental processing framework supports:

- Efficient batch processing
- Reduced pipeline execution time
- Controlled rerunnable execution
- Idempotent data loading
- Reliable historical tracking
- Scalable processing for growing retail datasets

This design enables production-oriented data processing while maintaining consistency, reliability, and operational scalability across the pipeline.



Customers Pipeline Demonstrating Incremental Load Logic Using Metadata and Lookup Activities in Azure Data Factory

```
1: READ CONTROL TABLE

jdbc_url = "jdbc:sqlserver://customerretailproject.database.windows.net:1433;database=Bretailcustomers"

properties = {
    "user": "username",
    "password": "Acme@123",
    "driver": "com.microsoft.sqlserver.jdbc.SQLServerDriver"
}

control_df = spark.read.jdbc(
    url=jdbc_url,
    table="(SELECT * FROM dbo.control_table WHERE source_name = 'Inventory') as t",
    properties=properties
)

control_df.display()
control_df.printSchema()
```

```
2: GETTING LAST WATERMARK

last_wm = control_df.select("last_watermark_value").first()[0]
print(last_wm)
```

2020-05-06 21:23:29.997000

```
3: READ BRONZE

bronze_df = spark.read.format("parquet").load( "abfss://bronze@strretail1000.df..."
```

Notebook using Incremental Logic Using Last watermark

```
7: COUNT

total_count = clean_df.count()
silver_count = silver_df.count()
rejected_count = rejected_df.count()

print(f"Total: {total_count}")
print(f"Inserted (Silver): {silver_count}")
print(f"Rejected: {rejected_count}")
```

(9) Spark Jobs

Total: 25
Inserted (Silver): 25
Rejected: 0

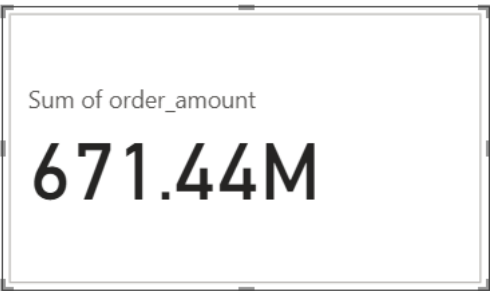
Using count to check rows before and after

POWER BI REPORTING

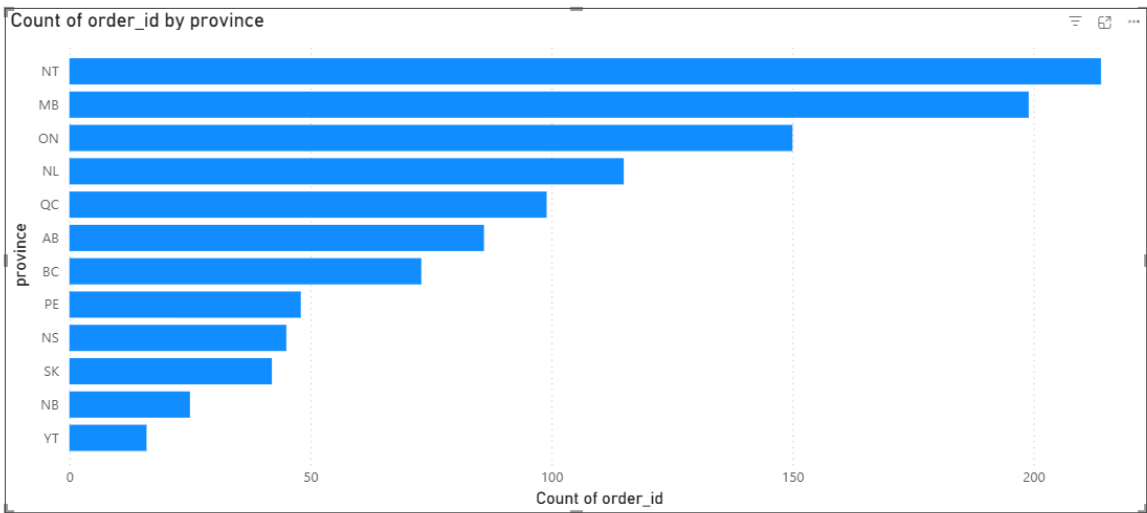
The Gold layer datasets are connected to Power BI to create interactive dashboards and business reports. Dimensional and fact tables are used to build analytics-ready visualizations such as Total Sales, Customer Metrics, Sales by Province, Monthly Sales Trends, and Top Store Performance.

The reporting layer follows a star-schema model using dimension and fact relationships to support efficient filtering, aggregation, and business analysis. Power BI dashboards provide business insights through KPI cards, trend analysis, and interactive visual reporting.

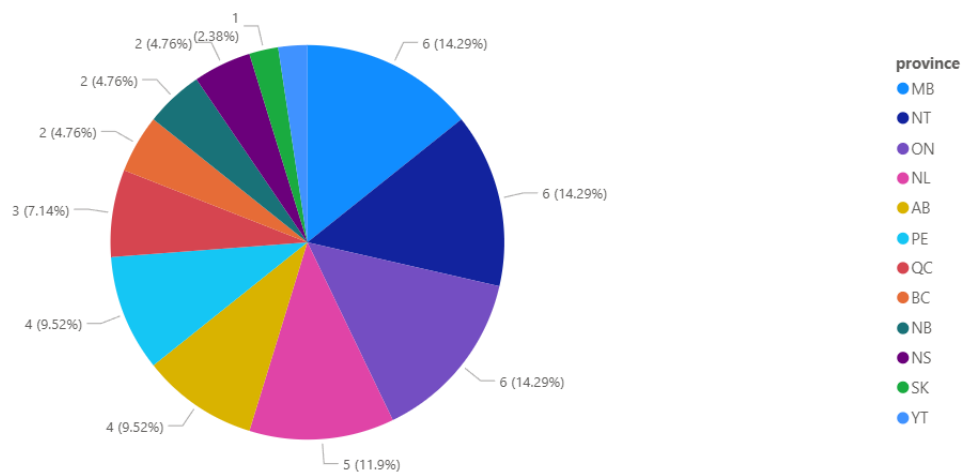
Total Sales



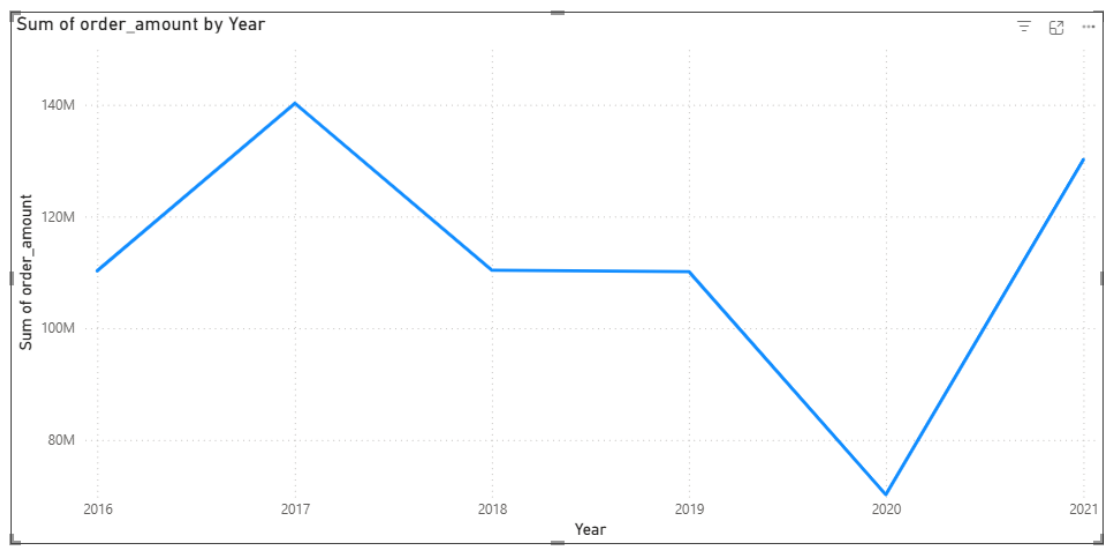
Sales Distribution by Province



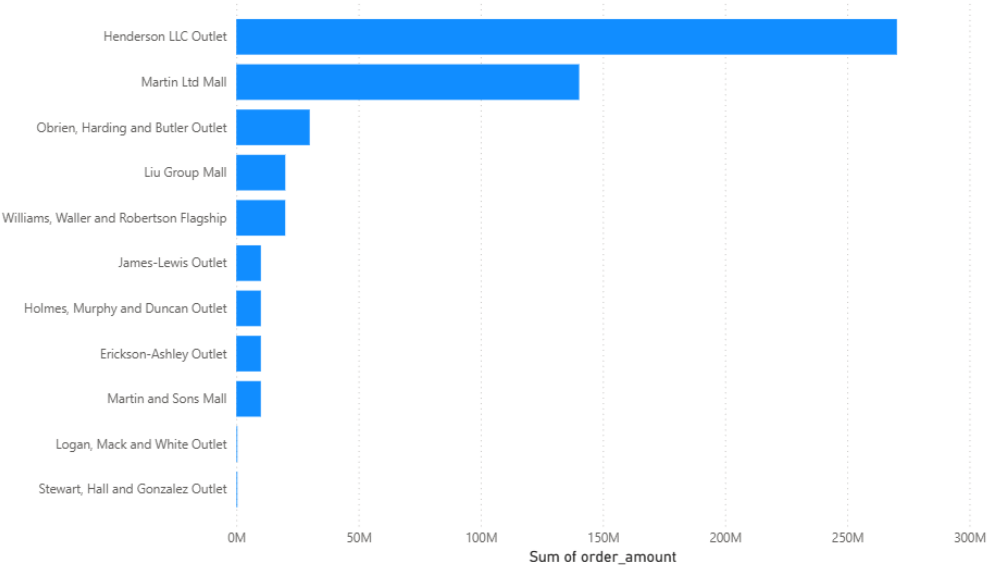
Top Customers by Province



Monthly Sales Trend



Top Stores by Revenue



ERROR HANDLING & REJECTED RECORD PROCESSING

The pipeline implements validation and error-handling mechanisms to identify invalid, incomplete, and malformed records during processing. Records failing validation checks such as invalid email formats, null mandatory fields, malformed dates, and incorrect categorical values are separated into rejected datasets for auditability and troubleshooting.

Rejected records are stored independently from validated business-ready datasets to maintain data quality and prevent corrupted data from entering downstream analytics layers. The pipeline also supports graceful handling of no-data scenarios and rerunnable execution without causing pipeline failures.

Just now (2s)12

```
rejected_path = "abfss://silver@strretail360.dfs.core.windows.net/inventory/inventory_rejected"

rejected_df = spark.read.format("delta").load(rejected_path)

rejected_df.display()
```

(3) Spark Jobs

rejected_df: pyspark.sql.connect.dataframe.DataFrame = [inventory_id: string, product_id: string ... 5 more fields]

Table +

	inventory_id	product_id	store_id	available_quantity	last_restock_date	inventory_status	ingestion_time
1	INV_64656	null	STR-26-000001	21	2022-02-05T00:00:00.000+00:...	null	2026-05-07T02:31:06.831+00:...
2	INV_63817	null	STR-26-000002	36	null	in_stock	2026-05-07T02:31:06.831+00:...
3	INV_48360	null	STR-26-000003	424	null	null	2026-05-07T02:31:06.831+00:...
4	INV_25874	null	STR-26-000004	22	2025-09-11T00:00:00.000+00:...	in_stock	2026-05-07T02:31:06.831+00:...
5	INV_40406	null	STR-26-000005	283	2021-11-19T00:00:00.000+00:...	in_stock	2026-05-07T02:31:06.831+00:...
6	INV_11163	null	STR-26-000006	68	2020-12-05T00:00:00.000+00:...	in_stock	2026-05-07T02:31:06.831+00:...
7	INV_80035	null	STR-26-000007	72	null	in_stock	2026-05-07T02:31:06.831+00:...
8	INV_15383	null	STR-26-000008	104	null	null	2026-05-07T02:31:06.831+00:...
9	INV_69959	null	STR-26-000009	111	null	in_stock	2026-05-07T02:31:06.831+00:...
10	INV_89963	null	STR-26-000010	5	2025-02-03T00:00:00.000+00:...	null	2026-05-07T02:31:06.831+00:...
11	INV_57121	null	STR-26-000011	349	null	null	2026-05-07T02:31:06.831+00:...
12	INV_21728	null	STR-26-000012	77	2021-07-15T00:00:00.000+00:...	in_stock	2026-05-07T02:31:06.831+00:...
13	INV_40842	null	STR-26-000013	50	2021-02-03T00:00:00.000+00:...	in_stock	2026-05-07T02:31:06.831+00:...
14	INV_60368	null	STR-26-000014	112	2024-06-19T00:00:00.000+00:...	in_stock	2026-05-07T02:31:06.831+00:...
15	INV_76225	null	STR-26-000015	31	null	in_stock	2026-05-07T02:31:06.831+00:...
16	INV_64833	null	STR-26-000016	106	null	in_stock	2026-05-07T02:31:06.831+00:...
17	INV_95466	null	STR-26-000017	126	2020-11-01T00:00:00.000+00:...	null	2026-05-07T02:31:06.831+00:...
18	INV_13824	null	STR-26-000018	179	null	in_stock	2026-05-07T02:31:06.831+00:...
19	INV_59996	null	STR-26-000020	121	2020-02-03T00:00:00.000+00:...	in_stock	2026-05-07T02:31:06.831+00:...
20	INV_19895	PRD-26-000005	STR-26-000011	null	2025-03-23T00:00:00.000+00:...	in_stock	2026-05-07T02:27:08.895+00:...

38 rows | 2.49s runtime

Rejected Record Handling and Rejected Table Processing in Silver Layer

Microsoft Azure | Data Factory | dftailcustomer360

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines (16)
 - clean_tables (2)
 - pipeline1
 - pl_clean
 - master_pipeline (3)
 - master_trigger
 - pl_master
 - sub_bronze_to_silver
 - pipelines_architecture (11)
 - bronze_to_silver_transform (5)
 - pl_customers_bronze_to_silver
 - pl_inventory_bronze_to_silver
 - pl_products_bronze_to_silver
 - pl_sales_bronze_to_silver
 - pl_stores_bronze_to_silver
 - raw_to_bronze_pipelines (5)
 - pl_raw_to_bronze_customer
 - pl_raw_to_bronze_inventory
 - pl_raw_to_bronze_product
 - pl_raw_to_bronze_order
 - pl_raw_to_bronze_coupon

Activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Activities

- Execute Pipeline
- Wait

Execute Pipeline

exe_raw_to_bronze
pl_raw_to_bronze

Wait

wait

Execute Pipeline

exe_bronze_to_sil...
pl_bronze_to_silver

Execute Pipeline

exe_pl_gold_aggr...
pl_gold_aggregate

Parameters Variables Settings Output

Pipeline run ID: 000436c3-3b5e-4820-967b-4b90ac87c185 Pipeline status: Succeeded

View debug run consumption

Monitor in Azure Metrics Export to CSV

Showing 1 - 4 of 4 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
exe_pl_gold_aggregate	Succeeded	Execute Pipeline	5/6/2026, 5:43:07 PM	3m 9s			7d9d63df-5500-4850-9858-82e66
exe_bronze_to_silver	Succeeded	Execute Pipeline	5/6/2026, 5:26:52 PM	16m 16s			377a88ef-d6ec-46f3-a002-a05407
wait	Succeeded	Wait	5/6/2026, 5:26:41 PM	11s			20d86585-a7d4-4708-99dd-784b1
exe_raw_to_bronze	Succeeded	Execute Pipeline	5/6/2026, 5:10:45 PM	15m 56s			65258ca7-bcb3-4d9c-b94c-a87ae1

Master Pipeline Orchestrating End-to-End Retail Data Processing

CONCLUSION

The Retail Customer 360 Data Engineering Pipeline successfully demonstrates an end-to-end cloud-based data engineering solution using Azure technologies. The project implements scalable ingestion, incremental processing, data validation, transformation, Delta Lake merge operations, and Medallion Architecture principles to process retail datasets efficiently and reliably.

The pipeline supports rerunnable and idempotent execution while maintaining data quality through cleansing, deduplication, and rejected-record handling. The final Gold-layer datasets are transformed into analytics-ready reporting tables that support Power BI dashboards and business intelligence reporting.

Overall, the project showcases practical implementation of modern data engineering concepts including cloud storage, distributed processing, incremental data loading, dimensional modeling, and enterprise-style reporting workflows within a production-oriented architecture.